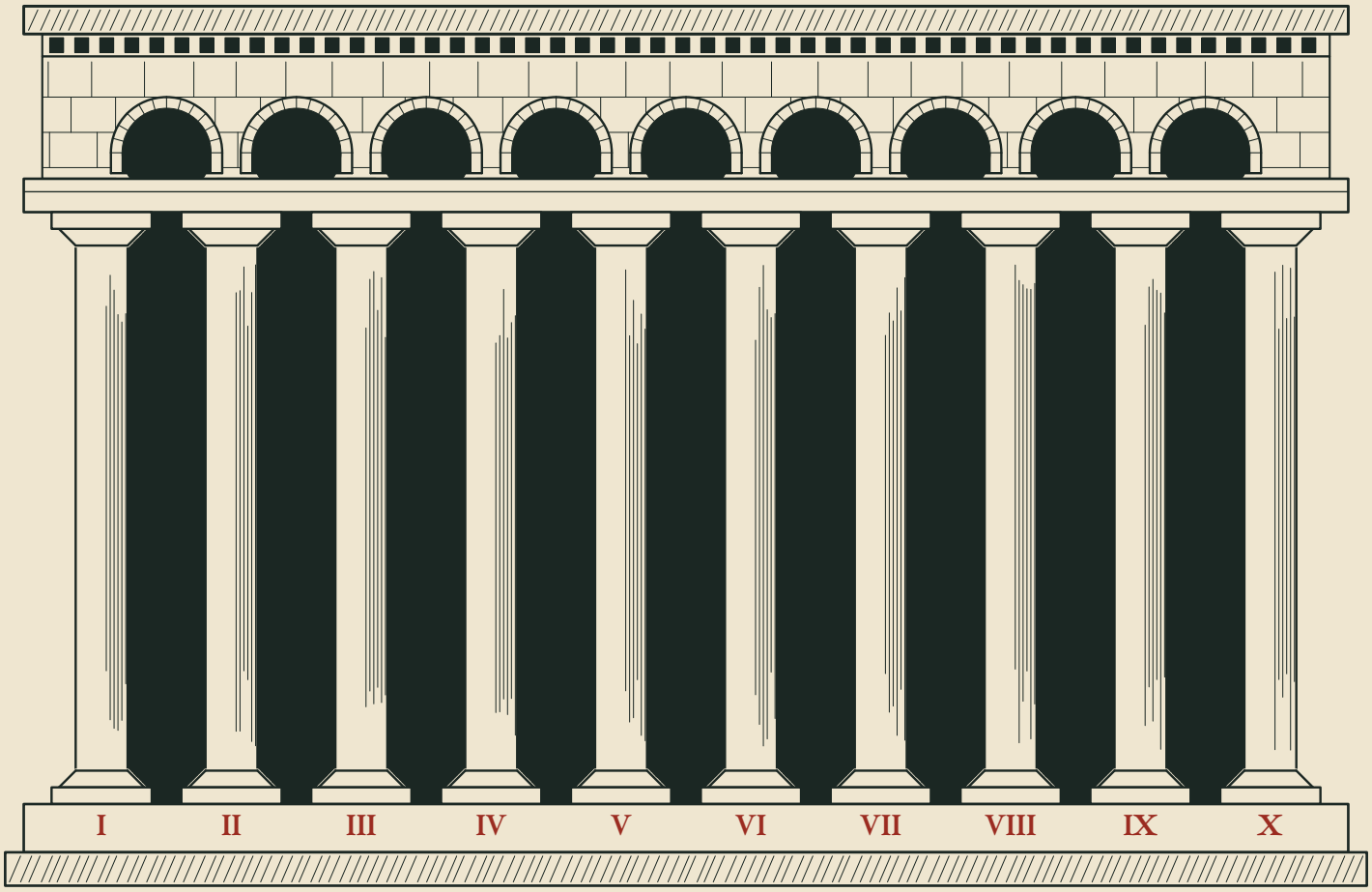
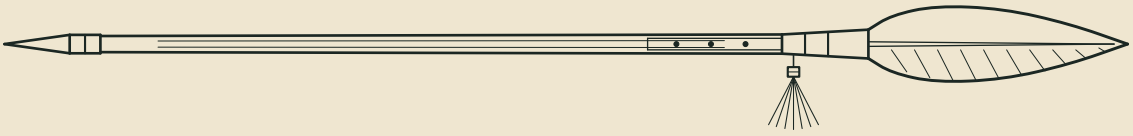
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The Ten Pillars

of Tradecraft





Sebastien Arseneault

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Entropy

Time tells all, and time takes all. Every system decays toward disorder. Hardware ages. Software rots. What was secure on Monday is a vulnerability by Friday. Vendor support lapses, new techniques arrive, old assumptions break. Nothing built on the surface will last.

And entropy does not spare the operator. The same law that rots the target's defenses rots your own work. The tool that slipped past every sensor last year is signatured today. The technique that was yours alone is public by autumn. The access you fought for decays the moment you win it, patched and rotated and hardened out from under you. The operator who lives by technique is always running, and the clock he runs against never stops. In the end it always wins.

So do not build there. What follows are not techniques, for techniques are the first thing entropy claims. They are the disciplines beneath technique. A tool ages; the judgment to know when not to use it does not. An exploit is patched; the habit of confirming before you touch is not. Position is taken back; the instinct to seek position first is permanent. Build on these, because they were never made of anything time can reach.

I

Access

The name of the game is access. Everything we do rests on reaching a place we are not meant to reach, and holding it once we have. Usually we are denied that privilege, and the exploit is only one way past the denial. It is rarely the best one.

Most access is walked into, not broken into. A password reused from an old breach. A share left open. A key resting beside the thing it unlocks. The beginner studies his target, finds the exploit that matches, and reaches for it first, when the quieter path was open all along. Measure yourself by the access you gained, not by how impressive the tool was that gained it.

But there is a harder kind. When there is no open door, no reused key, no misconfiguration, access must be made. You build it out of the target's own code, from a flaw the author never saw. This is vulnerability research: reading the binary with no source, learning how the thing works and how it fails. And it is the zero-day: turning that flaw into access that holds against a target built to deny it. The walked-in door and the manufactured one are the same door in the end. Both are judged the same way.

For access is not one thing but three. Is it flexible, or does it do only what you first made it do? Can you extend it, insert your own capability, bend it to a purpose you had not yet imagined? And can you keep it, quietly, against a defender who would take it back? A foothold that does one fixed thing is a payload. A foothold you can shape at will, and hold, is access.

II

Certainty

Success is difficult to define, and there is never one answer for what it looks like. It may seem as simple as a 200 response. But a clean “not vulnerable” is success too. So is a tool that stops because you built it to stop. The only true failure is the undefined state: the operator makes contact and reaches no clear decision, acts and cannot say what happened. Success is not whether you won. It is whether you know what you or your tool did.

It has a familiar shape. The shell that came back once and never again. The scan that returned half the hosts, and you cannot say which half. The payload that ran, or did not, where the only evidence is that nothing broke. Each of these looks like progress. None of it is ground you hold.

This is the discipline that separates the operator from the passenger. Getting a shell you cannot explain is more dangerous than getting nothing, because you will repeat something you do not understand until it fails against a target that matters. Certainty is control of the ground. Uncertainty, not failure, is the enemy. A clean “no” is a win. A murky “yes” is a liability. After every action, be able to finish one sentence: what you expected, what happened, and whether you know why. When you cannot finish it, that gap is the thing to close before anything else.

III

Protection

The tool is worth more than the target. This is the first inversion the beginner must learn, because every instinct runs the other way. The target is in front of you, urgent and real. The tool is only a means, and means feel spendable. But the target is one, and the tool is many. Spend the tool to take the target, and you have traded the many for the one.

A capability that is used is a capability that can be seen. A capability that is seen is studied, and a capability that is studied is dead. So the tool that stays silent is worth years, and the tool that is spent is worth a single afternoon. The value was never in using it. The value was in still having it.

From this one truth the whole discipline follows. You do not carry one capability, you carry tiers, and you spend the cheapest that will do the work. The old, public exploit before the one nobody has seen. The commodity tool before the one you wrote. You hold the best in reserve for the target that alone could justify losing it. And when the choice is between the prize and the tool, you learn to walk away from the prize, because the operator who protects his means outlasts the operator who spends them. Restraint is not caution. It is arithmetic.

IV

Patience

The probe that finds the door is often the probe that closes it. This is the operator's oldest paradox: to know a thing is weak, you are tempted to test its weakness, and the test is the very act that betrays you. The knock confirms the lock, and it also wakes the house.

So the discipline is to learn without touching. Everything a target does, it does in a way that reveals it. How it answers ordinary questions, how long it takes, what it says of itself without meaning to. The patient operator reads these and knows what he needs to know, while the impatient one reaches out, confirms his suspicion, and finds that confirming it was the same as announcing himself.

Patience is not waiting. It is the refusal to spend certainty you have not yet secured. The one interaction that matters must be the one that succeeds, because a failed first touch does not simply miss. It burns the tool, the access, and the method all at once, and it warns the defender that someone is at the door. The first touch is often the only touch you get. Everything before it is watching, and watching costs nothing.

V

Restraint

More is the instinct that gets you caught. Every foothold you take is a thing that can be found. Every action you repeat is a pattern that can be seen. The operator who spreads himself across a network to be safe has done the opposite of safe, because he has multiplied the surfaces on which he can be discovered.

The defender does not watch for permission. He watches for volume, for repetition, for the shape of someone who is everywhere at once. Nothing you are allowed to do protects you if the doing of it, at scale, draws the eye. And so the small footprint is not modesty. It is survival.

Hold only what you need. One beacon that answers is worth more than five that might be noticed. Return to nothing you already hold. Set your limits before you begin, when your judgment is clean, so that in the moment, when the temptation is to take one more, you have already decided not to. Restraint is a decision made in advance against a self you cannot yet see.

VI

Memory

To fail once is a cost. To fail twice the same way is a choice. The difference between them is memory, and memory is the one advantage that compounds while everything else decays.

Most operators remember only what wounded them, the dramatic failures, and forget the quiet successes entirely. But a record of only failure has no measure. You cannot tell a reliable method from a lucky one if you never wrote down the times it simply worked. The truth of a technique lives in the whole account, the wins beside the losses, kept plainly and read again when the heat of the moment has passed.

And memory is only half the discipline. The other half is turning what you remember into something you can no longer forget, a rule that makes the old mistake impossible rather than merely unlikely. To record the burn is to have a diary. To make the burn structurally impossible is to have a discipline. Only the second one protects you, because only the second one does not depend on you remembering in the moment you are least able to.

VII

Separation

Separation is the refusal to let any one thing carry the whole. It works on two fronts, and both are the same instinct wearing different clothes.

The first is the separation of building from doing. Building wants patience, and testing, and the willingness to be wrong slowly. Doing wants speed, and commitment, and a window that is closing now. Put them in the same room and speed always wins, and speed will make you ship the thing you have not tested because the moment demanded it. So freeze your work before you begin, and enter the operation with what is finished. When something fails in the moment, you adapt with what you have. You do not build. The builder and the operator are the same person wearing different discipline, and the art is knowing which one you are right now.

The second is the separation of the parts. No single piece should know the whole, and no single loss should cost you everything. The tool must not carry the target within it. The path in must not be the path out. What you build must be arranged so that whoever finds one thread cannot pull the rest of the cloth toward him. This is why the strongest work is assembled from pieces that mean nothing alone, so that a defender holding one of them holds a fragment that points nowhere. To compartment is to lose gracefully, to give up a piece without surrendering the position. The operator who ties everything to everything has built a machine that fails all at once. The one who separates has built a thing that can bleed a little and live.

VIII

Provenance

Memory keeps the outcomes. Provenance keeps the changes, and you cannot fix what you cannot trace. Every change you make to your craft is a change that may someday betray you, and when it does, you will need to know which one it was. The operator who edits blindly, keeping no record of what he altered or why, is helpless the day his work begins to fail, because he cannot walk backward to the moment it broke.

This is why your configuration belongs nowhere but in code, kept as code is kept: every change recorded, attributed, carrying the reason it was made. Not a pile of edits, but a history you can read and, when you must, reverse. When a technique that worked begins to fail, you do not theorize about why. You walk the history back, change by change, to the single one that broke it. The operator with a record bisects to the cause in an hour. The one without it guesses for a week and often guesses wrong.

The virtue looks like bureaucracy and is its opposite. To account for every change is to keep the power to unwind, to know at any moment what you hold, what you altered, and what you can pull back. This is also how you avoid losing everything at once. The arsenal whose history is lost cannot be recalled when it turns dangerous, and a capability you can no longer control is a capability that now controls you. Keep the account. Let every change carry its reason. Not so you may answer to anyone, but so that when the ground shifts under you, you can find the exact stone that moved, and set it back.

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IX

Position

Where you stand matters more than what you carry. This is the truth that humbles the toolmaker and elevates the operator, and it is the hardest of the ten to accept, because it means your cleverness is worth less than your ground.

Advantage lives in two places only: in the quality of your tools, and in the position from which you use them. Of the two, position is the greater, because the crude tool from commanding ground defeats the fine tool from weak ground every time. The most powerful systems ever built were poorly made and dominated anyway, for one reason. They stood where everything had to pass. No elegance overcomes a bad position, and no amount of skill talks its way past ground it does not hold.

Learn the ceiling early, because it will save you years of the wrong ambition. There is a position you will never reach. The one who sits astride the wire itself, who sees all that flows because everything flows through him, holds ground that no tool you could build will ever take and no sum you could raise will ever buy. That position is not won by cleverness. It is held by force and law and reach you do not have. To spend your life trying to out-tool it is to lose slowly to arithmetic.

So do not fight for the ground you cannot hold. Find the ground that is yours. Position cannot be bought, only occupied, and there is always a place near you that no rival has bothered to take. The deep knowledge of one kind of terrain. The trust that took years to earn. The quiet corner of a field the giants overlook because it is beneath them. Capability is common now, and everyone has good tools. Position is rare, because it must be stood upon and kept. When you are stuck, do not ask what better tool you need. Ask what ground you could hold that no one else has thought to stand on.

X

Invisibility

The finest work leaves no mark. To survive on the ground you have taken, you must become nothing to the eye that looks for you, meaningless in isolation, silent under inspection, indistinguishable from the ordinary until the moment you choose otherwise.

This is a way of building, not a trick added at the end. The invisible thing is designed backward from the eye that hunts it, made from the beginning to mean nothing on its own. What takes without subtracting is the hardest of all to catch, because it triggers no alarm, draws no power, leaves the door as it found it. The strongest presence is the one that cannot be told from absence.

So build for the moment of inspection, not the moment of use. Ask of everything you leave behind: if it were looked at now, doing nothing, would it give itself away? Refine it until the answer is no. The loud operator wins the engagement and loses the war. The quiet one is never known to have been there at all.

Tempo

Look back over the ten, and you will find they are all the same shape. Each is a thing decided in the calm, held ready against the one force that undoes operations: the pressure to act before you have thought.

That pressure is the true enemy. Not the defender, not the clock, not the target. The enemy is the moment that says move now, use the best tool, take the extra host, skip the record, build the fix where you stand. Every failure in this craft can be traced back to a good decision surrendered to a fast one. The defender does not beat the operator. The operator's own haste does, and the defender merely arrives to collect.

This is why the professional is not faster than the rest. He is slower where it counts and instant where it matters, and the difference between the two is everything he decided in advance. Access, certainty, protection, patience, restraint, memory, separation, provenance, position, invisibility. Ten disciplines, and every one of them is a brake set before the descent, so that when the road drops away beneath him he does not have to find the pedal in the dark.

The techniques will not last. They never do. The tool that works today is signatured tomorrow, and the door that stands open now will be sealed by someone who reads the same notes you do. Build your craft on technique and you build on sand that the tide is already coming for. But the disciplines beneath technique were never made of anything time can reach. They held in the age of dead drops and they hold now, because they are not methods but the refusal to be hurried into abandoning them.

Decide in the calm. Hold in the storm. When the moment comes, there will be nothing left to decide, and that stillness, not speed, is the whole of the art.